# Business Finance Midterm Cheat Sheet: Ch.2–7

Formulas · BA II Plus keystrokes · Key concepts | Prof. Gemma Lee, Kyung Hee Univ. | Lecture-slide scope only

## 0. BA II Plus Setup

### 0.1 Reset (before every problem)

- `2nd CLR TVM` clears N, I/Y, PV, PMT, FV (CE|C, ON/OFF, 2nd QUIT do **not**)
- Cash-flow (NPV) problems: `2nd CLR WORK`
- P/Y = 1: `2nd P/Y` `1 ENTER` `↓` `1 ENTER` `2nd QUIT`
- 4 decimals: `2nd FORMAT` `4 ENTER` `2nd QUIT`
- BGN on screen → switch to END (coupons and ordinary annuities pay at period end)
- Full reset (`2nd RESET ENTER`) sets P/Y back to 12 → reset P/Y = 1

### 0.2 Input rules

- Enter 4 of N, I/Y, PV, PMT, FV; `CPT` + the 5th
- Cash out = **−**, cash in = **+**; same sign on PV and FV → Error 5
- I/Y = rate per period as a % number (1% → 1)
- Single sum → PMT = 0
- Negative answer = amount to pay now, not an error
- Answer off by exactly (1 + r) → BGN mode is on

## Ch.2 Financial Statements, Taxes and Cash Flow

### 2.1 Balance Sheet

- Snapshot at a point in time; **Assets = Liabilities + Equity**
- Assets in decreasing liquidity; liabilities by when due
- NWC = current assets − current liabilities
- Liquidity = speed and ease of conversion to cash; avoids financial distress

### 2.2 Book vs. Market Value; Equity Accounts

- Book = accounting value (historical cost); market = price in actual trade → use market value for decisions
- Common stock = shares × par; paid-in surplus = (issue price − par) × shares
- Equity = paid-in capital + retained earnings; par = arbitrary nominal amount
- Ex: 100 sh, par $1, issued at $6 → $100 + $500 = **$600**

### 2.3 Income Statement

- Performance over a period; sales first, then subtract costs
- Sales − COGS − Depreciation = **EBIT**; EBIT − Interest = taxable income; − Taxes = **NI**
- NI = dividends + addition to retained earnings
- Ex: 1,509 − 750 − 89 = 670; 670 − 70 = 600; tax 21% = 126; NI = **474**
- Trap: interest is deducted **after** EBIT

### 2.4 GAAP vs. IFRS

- GAAP: FASB, rules-based | IFRS: IASB, principles-based | Korea: K-IFRS
- Same purpose; presentation order flipped (GAAP current items first, IFRS non-current first)

### 2.5 Taxes

- Average = total tax ÷ taxable income | Marginal = rate on the next $1 (use for decisions)
- Flat tax → marginal = average; progressive → marginal > average
- US flat 21% (2017 reform; before: progressive up to 35%) | Korea progressive 10/20/22/25%
- OECD Pillar Two: 15% minimum effective tax for large MNEs (≥ €750M)
- Ex: Korea ₩500M: 10% × 200M + 20% × 300M = ₩80M → average **16%**, marginal **20%**

### 2.6 Cash Flow (CFFA)

- **CFFA = OCF − NCS − ΔNWC = CF to creditors + CF to stockholders**
- OCF = EBIT + Depreciation − Taxes

- NCS = End net fixed assets − Beg net fixed assets + Depreciation
- ΔNWC = End NWC − Beg NWC
- CF/CR = Interest − Net new borrowing | CF/SH = Dividends − Net new equity
- Ex: OCF = 670 + 89 − 126 = 633 | NCS = 1,709 − 1,644 + 89 = 154 | ΔNWC = (1,403 − 389) − (1,112 − 428) = 330
- Ex: CFFA = 633 − 154 − 330 = **149** | check: CF/CR 70 − 46 = 24, CF/SH 165 − 40 = 125, sum 149
- Ex (Zephyr): 3,180 − 1,400 − 600 = **1,180**
- Traps: depreciation is added back (OCF, NCS) | current-liability increase lowers NWC | negative CF/CR = borrowing > interest paid | omit any component → wrong
- Cash flow ≠ accounting statement of cash flows

## Ch.3 Working with Financial Statements

### 3.1 Common-Size Statements; Users

- Balance sheet ÷ total assets | income statement ÷ sales; compare across years and across different-size firms
- Ex: inventory 84,600 ÷ 470,000 = **18.0%**
- Internal users: evaluation, planning, control | External: lenders, suppliers, customers, investors, competitors / acquirers / regulators

### 3.2 Ratio Categories

- Liquidity (short-term solvency) | Leverage (long-term solvency) | Turnover (asset management) | Profitability | Market value
- Slide firm: Sales 2,361; NI 474; TA 3,630; TE 2,625; price $115; 33M shares

### 3.3 Liquidity

- Current = CA ÷ CL → **1.29**
- Quick = (CA − Inventory) ÷ CL → **0.52** (excludes inventory)
- Cash = Cash ÷ CL → **0.18**

### 3.4 Leverage and Coverage

- Total debt ratio = (TA − TE) ÷ TA → **0.28**
- D/E = TD ÷ TE → **0.38** | EM = TA ÷ TE = 1 + D/E → **1.38**
- TIE = EBIT ÷ Interest = 741 ÷ 141 → **5.26**
- Cash coverage = (EBIT + Dep.) ÷ Interest = 1,017 ÷ 141 → **7.21** (always ≥ TIE)

### 3.5 Turnover

- Inventory = **COGS** ÷ Inventory → 3.18 (numerator is COGS, not sales)
- Receivables = Sales ÷ AR → 12.56 | Payables = COGS ÷ AP → 3.91
- Total asset turnover (TAT) = Sales ÷ TA → 0.65

### 3.6 Profitability

- PM = NI ÷ Sales → 20.08% | ROA = NI ÷ TA → 13.06% | ROE = NI ÷ TE → 18.06%
- Same numerator; only the denominators differ

### 3.7 Market Value

- EPS = NI ÷ shares → 14.36 | P/E = price ÷ EPS → 8.01
- P/S = price ÷ (Sales/share) → 1.61 | M/B = price ÷ (TE/share) → 1.45 (book value per share)
- EV = market cap + debt − cash; EBITDA = EBIT + Dep.
- Ex: EV = 3,795 + 1,005 − 98 = 4,702; EBITDA = 1,017; **EV/EBITDA = 4.62**

### 3.8 DuPont

- **ROE = PM × TAT × EM** (a product, not a sum) | PM = profitability / cost control, TAT = asset efficiency, EM = leverage
- Ex: Coca-Cola 27.34% × 0.46 × 3.26 ≈ **40.7%** (slide value; inputs rounded)
- KO vs PEP: both ROE ≈ 40%, different paths
  - KO: sells concentrate, asset-light, high margin, low turnover
  - PEP: finished goods and snacks, ~2× sales, thin margin, high turnover and leverage
  - PEP current ratio < 1 is not a red flag (cash sales, long payables terms, strong credit)

### 3.9 Benchmarks and Limits

- Benchmarks: time-trend analysis; peer group (SIC / NAICS)
- Limits: no underlying theory, conglomerates, global (IFRS vs GAAP), accounting methods (FIFO / LIFO, depreciation), different year-ends, capital structure, seasonality / one-offs, window dressing

## Ch.4 Time Value of Money

### 4.1 Concepts

- A dollar today > a dollar tomorrow (it can be invested)
- Compounding (interest on interest) ↔ discounting
- Simple = interest on principal only | compound = on principal + past interest; gap grows with time
- Ex: $1,000, 10%, 3 yrs: simple 1,000 + 3 × 100 = 1,300 vs compound 1,000 × $1.10^3$ = **1,331** (interest-on-interest $31)

### 4.2 Formulas

- $FV = PV(1 + r)^t$ | $PV = FV \div (1 + r)^t$
- $r = (FV \div PV)^{1/t} - 1$ | $t = \ln(FV \div PV) \div \ln(1 + r)$
- Slides favor `CPT I/Y` and `CPT N` over memorizing r and t formulas

### 4.3 Keystrokes

- FV: $2,000, 8%, 5 yrs → `5 N` `8 I/Y` `0 PMT` `2000 +/− PV` `CPT FV` = **2,938.66**
- PV: $10,000 in 6 yrs, 7% → `6 N` `7 I/Y` `0 PMT` `10000 FV` `CPT PV` = **−6,663.42**
- r: $5,000 → $10,000 in 9 yrs → `9 N` `5000 +/− PV` `0 PMT` `10000 FV` `CPT I/Y` = **8.01%**
- t: $1,000 → $2,500 at 9% → `9 I/Y` `1000 +/− PV` `0 PMT` `2500 FV` `CPT N` = **10.63 yrs**
- Practice (a) FV of $8,000, 6.5%, 10 yrs → 15,017.10
- Practice (b) PV of $20,000 in 10 yrs at 6.5% → −10,654.52
- Practice (c) $8,000 → $20,000 in 10 yrs, r → 9.60%
- Practice (d) $8,000 → $20,000 at 6.5%, t → 14.55 yrs
- Trap: same sign on PV and FV → Error 5

## Ch.5 Discounted Cash Flow Valuation

### 5.1 Multiple Cash Flows

- Discount / compound each cash flow at its own t, then **add** (never average); including CF0 → NPV
- FV: $4,000 now + $7,000 in 1 yr, 8%, value at year 3
  - `3 N` `8 I/Y` `0 PMT` `4000 +/− PV` `CPT FV` = 5,038.85
  - `2 N` `7000 +/− PV` `CPT FV` = 8,164.80 → sum = **13,203.65**
- PV of uneven CFs 200 / 400 / 600 / 800 at 12%
  - `CF` `2nd CLR WORK` `0 ENTER ↓` (CF0) `200 ENTER ↓↓` `400 ENTER ↓↓` `600 ENTER ↓↓` `800 ENTER ↓↓`
  - `NPV` `12 ENTER ↓` `CPT` = **1,432.93**

### 5.2 Annuity Types and Formulas

- Annuity: level CFs, finite (car / home loans) | Annuity due: paid at start (rent, lease, insurance) → × (1 + r)
- Perpetuity: C ÷ r (preferred stock, consol) | Growing perpetuity: $C_1 \div (r - g)$, r > g
- $PV = C \times [1 - 1/(1 + r)^t] \div r$ | $FV = C \times [(1 + r)^t - 1] \div r$
- r = rate **per payment period** (not APR); t = **number of payments** (not years)

### 5.3 Annuity Keystrokes

- PV (loan amount): $632 / mo, 1% / mo, 48 mo → `48 N` `1 I/Y` `632 +/− PMT` `0 FV` `CPT PV` = **23,999.54**
- Practice: $450, 0.75%, 36 mo → 14,151.06
- PMT: $20,000, 0.5% / mo, 60 mo → `60 N` `0.5 I/Y` `20000 PV` `0 FV` `CPT PMT` = **−386.66**
- Practice: $15,000, APR 9% monthly, 4 yrs → r = 0.75%, N = 48 → −373.28
- r: cost $1,000, $200 / yr × 8 → `8 N` `1000 +/− PV` `200 PMT` `0 FV` `CPT I/Y` = **11.81%**
- Practice: $8,000, $1,200 / yr × 12 → 10.45%
- N: save $2,000 / yr at 7% to reach $50,000 → `7 I/Y` `2000 +/− PMT` `0 PV` `50000 FV` `CPT N` = **14.95 yrs**
- Practice: $400 / mo, 0.5% / mo, $30,000 → 63.85 mo (5.32 yrs)
- FV: $300 / mo, 0.75% / mo, 30 yrs → `360 N` `0.75 I/Y` `300 +/− PMT` `0 PV` `CPT FV` = **549,223.04**
- Practice: $2,500 / yr, 7%, 25 yrs → 158,122.59

### 5.4 Annuity Due

- Method 1: ordinary value × (1 + r) | Method 2: `2nd BGN` `2nd SET` `2nd QUIT`, enter as usual, then **restore END**
- Ex: $1,000 / mo, 1% / mo, 36 mo → ordinary PV 30,107.51 × 1.01 = **30,408.58**
- Practice: $1,800, 0.5%, 24 mo → 40,613.16 × 1.005 = 40,816.22
- Lease PMT (BGN): $48,000, 12% APR monthly, 4 yrs → `48 N` `1 I/Y` `48000 +/− PV` `0 FV` `CPT PMT` = **1,251.51** (= ordinary 1,264.02 ÷ 1.01)

### 5.5 Perpetuities

- PV = C ÷ r | Ex: preferred $1.50 / qtr at 2% / qtr → **75.00**
- Growing: $C_1 \div (r - g)$ | Ex: $2.00 next year, g 3%, r 10% → 2.00 ÷ 0.07 = **28.57**
- If $2.00 was just paid ($D_0$): $C_1 = D_0(1 + g)$ = 2.06 → 2.06 ÷ 0.07 = **29.43**

### 5.6 APR vs. EAR

- APR = quoted rate (periodic rate × m) | EAR = actual annual rate | **always compare in EAR**
- EAR = $(1 + APR/m)^m - 1$ | APR = $m \times [(1 + EAR)^{1/m} - 1]$ | continuous: EAR = $e^r - 1$
- More frequent compounding / higher rate → larger APR–EAR gap
- `2nd ICONV`: NOM → C/Y → `CPT EFF`; reverse: EFF → C/Y → `CPT NOM` (standard BA II Plus operation)
- Ex: 15% APR → annual 15.00%, semiannual 15.56%, quarterly 15.87%, monthly 16.08%, daily 16.18%
- Practice: 9% APR → quarterly 9.31%, monthly 9.38%, daily 9.42%
- Bank A 15% daily (EAR 16.18%) vs Bank B 15.5% annual (15.50%) → A for deposits
- Card A 22.9% monthly (EAR 25.46%) vs Card B 23.5% annual (23.50%) → B is cheaper
- EAR 12% as monthly APR → **11.39%** | EAR 9% as quarterly APR → **8.71%**
- Calculator returns a monthly rate: $250 / mo × 15 yrs → $75,000 → `180 N` `250 +/− PMT` `0 PV` `75000 FV` `CPT I/Y` = 0.5303% / mo → APR **6.36%**, EAR **6.55%**

### 5.7 Loan Types and Amortization

- Pure discount (T-bill, zero) | interest-only (corporate bond) | amortized (home / car loan)
- Loan balance = PV of remaining payments
- Payment constant; interest falls, principal rises
- Interest = beginning balance × r | principal = PMT − interest | ending = beginning − principal
- Ex: $5,000, 9%, 5 yrs → `5 N` `9 I/Y` `5000 PV` `0 FV` `CPT PMT` = **−1,285.46**
  - Year 1: interest 450.00, principal 835.46, ending balance 4,164.54
  - Total paid 6,427.30 = principal 5,000 + interest 1,427.30
- Practice: $10,000, 8%, 4 yrs → PMT 3,019.21 | yr-1 ending balance 7,780.79 | yr-2 interest 622.46 | balance after 2 yrs 5,384.05

## Ch.6 Interest Rates and Bond Valuation

### 6.1 Bond Anatomy and Keys

- Face / par = repaid at maturity | coupon = fixed interest | coupon rate = fixed at issue | maturity
- YTM = market required return, changes daily | after issue only the **price** changes
- Price = PV(coupon annuity) + PV(face)
- Keys: N = periods to maturity | I/Y = YTM per period | PMT = coupon per period | FV = face | PV = price
- PMT and FV positive → PV comes out negative (cash you pay); solving YTM → enter PV as negative

### 6.2 Par, Discount, Premium

- Coupon rate = YTM → **par**
- Coupon rate < YTM → **discount** (capital gain at maturity)
- Coupon rate > YTM → **premium** (capital loss at maturity)

### 6.3 Pricing Keystrokes

- Discount: $1,000, 8% coupon, 10 yrs, YTM 10% → `10 N` `10 I/Y` `80 PMT` `1000 FV` `CPT PV` = **−877.11**
- Premium: same bond, YTM 6% → `10 N` `6 I/Y` `80 PMT` `1000 FV` `CPT PV` = **−1,147.20**
- Semiannual: 9% coupon, 7 yrs, YTM 8% → N = 2 × yrs, I/Y = YTM ÷ 2, PMT = coupon ÷ 2
  - `14 N` `4 I/Y` `45 PMT` `1000 FV` `CPT PV` = **−1,052.82**
- Zero coupon: 5 yrs, YTM 6% semiannual → `10 N` `3 I/Y` `0 PMT` `1000 FV` `CPT PV` = **−744.09**

## 6.4 Yield to Maturity

- YTM = single discount rate that sets PV of promised CFs = price (the bond's IRR)
- Assumptions: hold to maturity | no default | coupons reinvested at YTM
- Annual: 8% coupon, 8 yrs, price 918.50 → `8 N` `918.50 +/− PV` `80 PMT` `1000 FV` `CPT I/Y` = **9.50%**
- Semiannual: 7% coupon, 10 yrs, price 900.29 → `20 N` `900.29 +/− PV` `35 PMT` `1000 FV` `CPT I/Y` = 4.25 → × 2 = **8.50%** (EAY 8.68%)
- Three yields (8% coupon, price 918.50): coupon rate 8.00% < current yield 80 ÷ 918.50 = 8.71% < YTM 9.50%

## 6.5 Interest Rate Risk

- Price and yield move **inversely** (promised CFs are fixed); price curve is convex
- Longer maturity and lower coupon → more sensitive; zero coupon is most sensitive
- Interest rate risk = price risk (not default risk)
- 30-yr bond, yield 10% → 12%: coupon 5% → −17.50% | 10% → −16.11% | 15% → −15.61%

## 6.6 Clean vs. Dirty Price

- Dirty (invoice) = clean (quoted) + accrued interest
- Accrued = coupon × (elapsed ÷ coupon period)
- Ex: semiannual $40 coupon, 2 months elapsed, quote $985 → 40 × 2/6 = 13.33 → dirty **998.33**

## 6.7 Fisher Effect and Yield Components

- **1 + R = (1 + r)(1 + h)** | Ex: nominal 15.5%, inflation 5% → 1.155 ÷ 1.05 − 1 = **10.00%** real (R − h = 10.5% is only approximate)
- Nominal yield = real rate + inflation premium + interest-rate-risk premium (rises with maturity) + bond-specific: default, taxability, liquidity premiums
- Higher expected inflation → higher nominal yields, lower bond prices

## 6.8 Term Structure

- Relation between maturity and yield (pure-discount Treasuries); yield curve = its graph
- Upward sloping → expected inflation up | downward → expected inflation down

## 6.9 Treasuries and Municipals

- Treasury: no default risk, state / local tax-exempt, federally taxable
- Municipal: default risk, federally tax-exempt → lower yield, held by high-bracket investors
- Tax-equivalent yield = muni yield ÷ (1 − t) | 6 ÷ 0.65 = **9.23%** (t = 35%) | 6 ÷ 0.80 = **7.50%** (t = 20%)

## 6.10 Debt vs. Equity

- Creditors: no ownership, no vote | interest is a pre-tax (deductible) expense | dividends are paid after tax
- Unpaid interest → default; unpaid dividend → not default
- Liquidation: equity is the residual claim (last)

## 6.11 Indenture, Call, Covenants

- Indenture = contract between issuer and bondholders; trustee represents holders
- Terms: face, coupon, maturity | security | repayment (sinking fund) | seniority | call | covenants
- Call: issuer may repurchase before maturity | call premium = call price − face | deferred call = protected period
  - Exercised when rates fall → bond value lower; investors demand a higher yield
- Negative covenants: limit dividends, pledging assets, mergers, asset sales, extra long-term debt
- Positive covenants: maintain NWC, supply audited statements, maintain collateral, keep ratios
- Firms accept covenants because the yield reduction exceeds the flexibility cost

## 6.12 Ratings and Bond Types

- Investment grade AAA / Aaa to BBB / Baa | junk (speculative) BB / Ba and below
- BBB− / Baa3 = last investment grade (institutional-investment cutoff)
- Zero coupon: only CF at maturity | Floating rate: collar = floor + cap | Convertible: lower coupon
- Put provision: opposite of call, adds value | CoCo | Sukuk: asset-based return instead of interest

## 6.13 Bond Market

- Large volume but mostly **OTC** (no central exchange); dealer quotes; thin liquidity per issue; institution-dominated
- Transparency problem → TRACE (2002) reporting and disclosure

# Ch.7 Equity Markets and Stock Valuation

## 7.1 Stock vs. Bond Valuation

- Dividends at board discretion (coupons are contractual)
- No maturity → infinite cash flows
- Must estimate R and g; price is highly sensitive to both

## 7.2 Dividend Discount Model

- $P_0$ = PV of all future dividends
- "Sell later at a higher price" is already inside the model
- Non-dividend firms still have value only if they eventually pay
- No slide keystrokes for Ch.7 → solve with formulas

## 7.3 Four Growth Cases

- **Zero growth** (preferred): $P_0 = D \div R$
- **Constant growth**: $D_1 = D_0(1 + g)$; $P_0 = D_1 \div (R - g)$, requires $R > g$; $P_t = D_{t+1} \div (R - g)$
- **Nonconstant**: list $D_1 \ldots D_t$, find $P_t$, discount all; $P_t$ is discounted t periods
- **Two-stage** ($g_1$ for t yrs, then $g_2$): $P_0 = D_1/(R - g_1) \times [1 - ((1 + g_1)/(1 + R))^t] + P_t/(1 + R)^t$, with $P_t = D_{t+1}/(R - g_2)$ (standard form; slide formula is an image)
- Sustainable growth is low single digits
- Zero: D \$3.00, R 10% → **30.00**
- Constant: $D_0$ \$2.00, g 5%, R 12% → $D_1$ = 2.10; 2.10 ÷ 0.07 = **30.00**
- Future price: $D_6 = 2.00 \times 1.05^6$; $D_6 \div 0.07$ = **38.29**
- Nonconstant: $D_0$ \$1.00, 20% growth for 3 yrs then 5% forever, R 15%
  - $D_{1\ldots3}$ = 1.20, 1.44, 1.728 | $D_4$ = 1.8144 | $P_3$ = 1.8144 ÷ 0.10 = 18.144
  - $P_0 = 1.20/1.15 + 1.44/1.15^2 + 1.728/1.15^3 + 18.144/1.15^3$ = **15.20**
- No dividend for 4 yrs, \$0.50 at end of yr 5, g 4%, R 20% → $P_4 = 0.50 \div (0.20 - 0.04) = 3.125$; $\div 1.20^4$ = **1.51**

## 7.4 Return Decomposition

- **R = dividend yield ($D_1 \div P_0$) + capital gains yield (= g under constant growth)**
- Dividend yield is like a bond's current yield
- Ex: P \$20, $D_1$ \$1, g 10% → 5% + 10% = **15%**

## 7.5 Multiples

- P/E, price-sales: fallback when there is no dividend
- Benchmark = comparable firms or own history; "comparable" drives the answer
- Losses → P/E meaningless; P/S still works
- Price = EPS × benchmark P/E | Ex: 3.00 × 18 = **54.00**
- Price = sales per share × benchmark P/S | Ex: 10.00 × 2.5 = **25.00**

## 7.6 Common Stock Rights

- Elect directors; vote (mergers, charter amendments, new shares)
- Dividends only after board declaration | residual claim | limited liability (loss up to paid-in)
- Preemptive right: first claim on new shares, prevents dilution
- Proxy: delegated vote (management usually favored); proxy fight: rival solicits proxies
- Dual-class: Class B 10 votes vs Class A 1 (founder control); one-share-one-vote varies by law / listing rules

## 7.7 Dividends

- Discretionary (non-payment ≠ default)
- Paid from after-tax income (asymmetric to deductible interest)
- US corporate dividends-received deduction (DRD) → corporations prefer preferred stock

## 7.8 Preferred Stock

- Fixed dividend on par; paid before common, after creditors
- Cumulative: arrearages paid first | normally no vote (may gain vote in arrears)
- Looks like debt but is not: missed dividend ≠ bankruptcy trigger; dividends not tax-deductible; equity legally and in accounting

### 7.9 Stock Markets

- Primary (firm receives funds) vs secondary (firm receives nothing)
- Dealer: holds inventory, bears risk | broker: agent, earns commission
- Bid = dealer buys | ask = dealer sells | spread = dealer's compensation
- NYSE: hybrid (DMM, mostly electronic) | NASDAQ: dealer market, competing market makers, ECNs narrow spreads
- Quote fields: Last / Close, Net change, Bid / Ask, Volume, Dividend, Yield = dividend ÷ price, P/E = price ÷ EPS

## Exam Trap Checklist

### T.1 Calculator

- Same sign on PV and FV → Error 5; cash out −, cash in +
- `2nd CLR TVM` / `2nd CLR WORK` before each problem; P/Y = 1
- Match units of N and I/Y (semiannual: N = yrs × 2, I/Y = rate ÷ 2)
- First CF now → annuity due = ordinary × (1 + r); turn BGN off afterwards

### T.2 Statements and Ratios

- Interest deducted after EBIT | NCS adds depreciation | OCF adds depreciation back
- Inventory turnover uses COGS | quick ratio excludes inventory | EM = 1 + D/E | M/B uses book value per share
- DuPont is a product, not a sum

### T.3 Rates and Bonds

- APR given → convert to the periodic rate first; EAR = $(1 + APR/m)^m - 1$; compare in EAR
- After-tax interest = interest × (1 − T); dividends paid from after-tax income
- Price and YTM move inversely; coupon rate < YTM → discount; clean + accrued = dirty
- Tax-equivalent yield: **divide** by (1 − t) (6% × 0.65 = 3.90% is wrong)

### T.4 Stocks and Written Answers

- Constant growth numerator is $D_1$ ($D_0 \times (1 + g)$); valid only if R > g
- $P_3$ is discounted 3 periods, not 4; losses → use P/S
- Written answers: definition + direction (up / down) + reason in one line

# BUSINESS FINANCE — FINANCIAL RATIOS

Chapter 3 | One-page formula sheet · Use figures from the same fiscal period where possible

## 1. COMMON-SIZE STATEMENTS

| Ratio / purpose | Formula |
|---|---|
| Balance sheet account (%) | Account ÷ Total Assets × 100 |
| Income statement item (%) | Item ÷ Sales × 100 |

## 2. LIQUIDITY — short-term solvency

| Ratio / purpose | Formula |
|---|---|
| Current ratio | Current Assets (CA) ÷ Current Liabilities (CL) |
| Quick ratio | (CA – Inventory) ÷ CL |
| Cash ratio | Cash ÷ CL |

## 3. LEVERAGE / LONG-TERM SOLVENCY

| Ratio / purpose | Formula |
|---|---|
| Total debt ratio | (Total Assets – Total Equity) ÷ Total Assets = Total Debt ÷ TA |
| Debt-to-equity (D/E) | Total Debt ÷ Total Equity |
| Equity multiplier (EM) | Total Assets ÷ Total Equity = 1 + D/E |
| Times interest earned (TIE) | EBIT ÷ Interest |
| Cash coverage | (EBIT + Depreciation) ÷ Interest |

## 4. ASSET MANAGEMENT / TURNOVER

| Ratio / purpose | Formula |
|---|---|
| Inventory turnover | COGS ÷ Inventory |
| Receivables turnover | Sales ÷ Accounts Receivable (AR) |
| Payables turnover | COGS ÷ Accounts Payable (AP) |
| Total asset turnover (TAT) | Sales ÷ Total Assets |

## 5. PROFITABILITY

| Ratio / purpose | Formula |
|---|---|
| Profit margin (PM) | Net Income (NI) ÷ Sales |
| Return on assets (ROA) | NI ÷ Total Assets |
| Return on equity (ROE) | NI ÷ Total Equity |

## 6. MARKET VALUE

| Ratio / purpose | Formula |
|---|---|
| Earnings per share (EPS) | Net Income ÷ Shares Outstanding |
| P/E ratio | Price per Share ÷ EPS |
| Price/Sales (P/S) | Price per Share ÷ Sales per Share |
| Market-to-book (M/B) | Price per Share ÷ Book Value per Share |
| Enterprise value (EV) | Market Value of Equity + Book Value of Liabilities – Cash |
| EBITDA | EBIT + Depreciation & Amortization |
| EV/EBITDA | Enterprise Value ÷ EBITDA |

## 7. DUPONT IDENTITY — decompose ROE

| Ratio / purpose | Formula |
|---|---|
| Core identity | ROE = PM × TAT × EM |
| Expanded | (NI ÷ Sales) × (Sales ÷ TA) × (TA ÷ Equity) = NI ÷ Equity |

## QUICK INTERPRETATION & EXAM CHECKS

**Liquidity:** higher ratios usually mean more short-term coverage, but excess idle assets may reduce efficiency. • **Leverage:** more debt can magnify shareholder returns and risk; TIE / cash coverage measure ability to cover interest. • **Turnover:** higher turnover generally means more efficient asset use; interpret inventory turnover in industry context. • **Profitability:** PM = profit per sales dollar; ROA = profit relative to assets; ROE = profit relative to equity. • **Common-size:** balance sheet uses total assets as denominator; income statement uses sales. • **DuPont:** ROE can come from high margin, efficient asset turnover, or high leverage. • **Calculation:** convert percentages to % only at the end; ratios/turnover are often shown as “x” (e.g., 1.5x). • **Comparison cautions:** compare across time and similar firms; accounting methods, seasonality, fiscal year-end, one-offs, and window dressing can mislead.

# Business Finance Midterm

Multiple Choice · Chapters 2–7 · 70 questions

## Exhibit A · Harborview Outfitters, Inc.

*Used in questions 1–7*

Balance sheets, December 31 ($ millions)

| Item | 2024 | 2025 |
|---|---|---|
| Assets | | |
| Cash | 150 | 200 |
| Accounts receivable | 600 | 700 |
| Inventory | 850 | 900 |
| Total current assets | 1,600 | 1,800 |
| Net fixed assets | 4,400 | 4,600 |
| Total assets | 6,000 | 6,400 |
| Liabilities and owners' equity | | |
| Accounts payable | 520 | 560 |
| Notes payable | 280 | 240 |
| Total current liabilities | 800 | 800 |
| Long-term debt | 1,800 | 1,900 |
| Common stock and paid-in surplus | 1,400 | 1,500 |
| Retained earnings | 2,000 | 2,200 |
| Total owners' equity | 3,400 | 3,700 |
| Total liabilities and owners' equity | 6,000 | 6,400 |

2025 income statement ($ millions)

| Item | 2025 |
|---|---|
| Net sales | 4,800 |
| Cost of goods sold | 2,900 |
| Depreciation | 500 |
| Earnings before interest and taxes | 1,400 |
| Interest paid | 200 |
| Taxable income | 1,200 |
| Taxes (21%) | 252 |
| Net income | 948 |
| Dividends | 748 |
| Addition to retained earnings | 200 |

1. [Exhibit A] Harborview Outfitters, Inc. ($ millions). The firm has 100 million shares outstanding and the share price is $74. What are its P/E, price-sales and market-to-book ratios?
   a. P/E 9.48; P/S 1.54; M/B 2.00
   b. P/E 7.81; P/S 1.54; M/B 2.00
   c. P/E 7.81; P/S 0.65; M/B 0.50
   d. P/E 7.81; P/S 1.54; M/B 0.50

2. [Exhibit A] Harborview Outfitters, Inc. ($ millions). What are the firm's 2025 times interest earned and cash coverage ratios?
   a. TIE 9.50; cash coverage 7.00
   b. TIE 7.00; cash coverage 9.50
   c. TIE 7.00; cash coverage 4.50
   d. TIE 6.00; cash coverage 8.50

3. [Exhibit A] Harborview Outfitters, Inc. ($ millions). On the 2025 common-size statements, what percentages are shown for inventory and for net income?

a. Inventory 18.75%; net income 14.81%

b. Inventory 14.06%; net income 19.75%

c. Inventory 50.00%; net income 25.62%

d. Inventory 14.06%; net income 14.81%

4. [Exhibit A] Harborview Outfitters, Inc. ($ millions). The firm has 100 million shares at $74 each. Enterprise value uses the book value of all liabilities. What is its EV/EBITDA ratio?

a. 5.21

b. 5.32

c. 3.89

d. 7.07

5. [Exhibit A] Harborview Outfitters, Inc. ($ millions). The firm has 100 million shares outstanding. Comparable outdoor-retail firms trade at 12 times earnings. What is the estimated share price?

a. $113.76

b. $88.80

c. $57.60

d. $9.48

6. [Exhibit A] Harborview Outfitters, Inc. ($ millions). Using 2025 year-end balances, what are inventory turnover, receivables turnover and total asset turnover?

a. 3.22; 6.86; 0.75

b. 5.33; 6.86; 0.75

c. 3.22; 4.14; 1.33

d. 3.22; 6.86; 1.33

7. [Exhibit A] Harborview Outfitters, Inc. ($ millions). During 2025, which net effect on cash comes from the changes in accounts receivable and accounts payable alone?

a. $140 source of cash

b. $140 use of cash

c. $60 source of cash

d. $60 use of cash

8. Which statement about a bond that is already outstanding is correct?

a. The coupon rate is reset whenever market rates change

b. The coupon rate is fixed at issue, while the yield to maturity changes with the market every day

c. The yield to maturity is fixed at issue, while the coupon rate moves with the market

d. The face value rises when market rates fall

9. On a balance sheet prepared under US GAAP (the RWJ textbook format), how are the items ordered?

a. Assets in order of decreasing liquidity; liabilities in ascending order of when they are due
b. Assets in order of increasing liquidity; liabilities in descending order of when they are due
c. Non-current assets first, then current assets; non-current liabilities first
d. Assets and liabilities in order of size

10. Hanwoo Corp.'s balance sheet shows shareholders' equity of $80 million, while its 20 million shares trade at $12 each. Which statement is correct?

a. The market value of equity is $80 million
b. The market value of equity exceeds book value by $160 million
c. Book value is the better estimate of what shareholders would receive if they sold today
d. The difference implies an accounting error in the balance sheet

11. A bond's coupon rate is above the market yield, and the yield stays unchanged until maturity. What happens to its price over time?

a. It rises further above par as maturity approaches
b. It falls toward par and equals the face value at maturity
c. It stays at the same premium until the final coupon
d. It drops below par halfway through its life

12. You take a $250,000 mortgage for 30 years at 6%, compounded monthly. What is the monthly payment?

a. $1,513.52
b. $1,498.88
c. $8,994.73
d. $15,000.00

13. A stock sells for $50.00. Its next dividend will be $2.00, and dividends are expected to grow at 6% a year forever. What are its dividend yield, capital gains yield and required return?

a. 4.0%; 6.0%; 10.0%
b. 4.0%; 4.0%; 8.0%
c. 6.0%; 4.0%; 10.0%
d. 4.24%; 6.0%; 10.24%

14. Four bonds have the same yield to maturity. Which one's price is most sensitive to a change in interest rates?

a. A 2-year bond with a 10% coupon
b. A 30-year bond with a 10% coupon
c. A 2-year zero coupon bond
d. A 30-year zero coupon bond

15. Which feature shows that preferred stock is equity rather than debt?
    a. It pays a fixed periodic dividend
    b. It ranks ahead of common stock in liquidation
    c. It is often callable by the issuer
    d. Skipping a preferred dividend is not a default and cannot trigger bankruptcy

16. In computing cash flow from assets, how is an increase in net working capital during the year treated?
    a. It is subtracted, because the firm tied up additional cash in working capital
    b. It is added, because the firm holds more liquid assets
    c. It is ignored, because NWC is a balance-sheet item
    d. It is added to operating cash flow as a source of cash

17. A bond offers a nominal return of 8% for one year, and expected inflation is 3%. What is the real rate of return?
    a. 5.00%
    b. 4.85%
    c. 11.24%
    d. 5.15%

18. A $1,000 bond pays a 6% coupon semiannually ($30 per coupon). Four months have passed since the last coupon, and the bond is quoted at $1,012. What is the invoice (dirty) price?
    a. $1,032.00
    b. $1,022.00
    c. $1,012.00
    d. $1,042.00

19. You put $250 into an account at the end of every month for 20 years. The account pays 4.8%, compounded monthly. What will you have at the end?
    a. $100,418.76
    b. $97,126.75
    c. $100,820.43
    d. $60,000.00

20. Which statement about using financial leverage to raise ROE is correct?
    a. Higher leverage raises ROE without any effect on risk
    b. Higher leverage raises ROE only when assets earn more than the after-tax cost of debt, and it increases the risk borne by shareholders
    c. Leverage changes ROA but leaves ROE unchanged
    d. Leverage raises ROE only for firms with high profit margins

21. A loss-making firm has sales of $12.00 per share. Comparable firms trade at 1.8 times sales. What is the estimated share price?

a. $6.67
b. $21.60
c. $13.80
d. $0.00

22. An investor in the 25% tax bracket considers a municipal bond yielding 4.5%. What is its equivalent taxable yield?

a. 3.38%
b. 6.00%
c. 5.63%
d. 4.50%

23. A firm issues 5,000 shares of common stock with a par value of $1 at $15 per share. How much is recorded as additional paid-in capital (paid-in surplus)?

a. $75,000
b. $70,000
c. $5,000
d. $80,000

24. A preferred share pays $1.25 every quarter forever. Investors require 8%, compounded quarterly. What is the share worth?

a. $15.63
b. $250.00
c. $60.66
d. $62.50

25. A $36,000 machine can be leased for 36 payments of $1,150, each due at the beginning of the month. What annual rate, compounded monthly, does the lease charge?

a. 0.82%
b. 10.33%
c. 9.31%
d. 9.87%

26. Why is interest expense not deducted when computing operating cash flow?

a. Because interest is a noncash expense
b. Because interest is a financing cost paid to creditors, not an operating cost
c. Because interest is already included in depreciation
d. Because interest is not tax-deductible

27. Holding the quoted APR constant, what happens to the effective annual rate as compounding becomes more frequent?

a. It falls
b. It stays equal to the APR
c. It rises
d. It rises only if the APR exceeds 10%

28. A bond's coupon rate is above the current yield to maturity on comparable bonds. How does it sell?

a. At par
b. At a discount
c. At a premium
d. It cannot be determined without the maturity

29. Why does net income generally differ from operating cash flow?

a. Net income is measured before taxes, whereas OCF is measured after taxes
b. Net income subtracts noncash charges such as depreciation and also subtracts interest, a financing cost
c. Net income excludes cost of goods sold, whereas OCF includes it
d. Net income uses cash-basis accounting, whereas OCF uses accrual accounting

30. A firm reports sales of $850,000, costs of $610,000, depreciation of $95,000 and interest expense of $32,000. The tax rate is 21%. What is net income?

a. $113,000
b. $89,270
c. $114,550
d. $145,000

31. An investment pays $2,000 at the end of quarter 1, $3,000 at the end of quarter 2, nothing in quarter 3 and $4,000 at the end of quarter 4. The required return is 8%, compounded quarterly. What is it worth today?

a. $8,539.67
b. $8,613.58
c. $7,363.99
d. $9,000.00

32. A $1,000 bond with a 7% annual coupon and 10 years to maturity is priced at $1,075, which implies a YTM of 5.98%. What is its current yield, and how do the three yield measures rank?

a. Current yield 6.51%; YTM < current yield < coupon rate
b. Current yield 6.51%; coupon rate < current yield < YTM
c. Current yield 7.53%; YTM < coupon rate < current yield
d. Current yield 7.00%; all three are equal

33. On a common-size balance sheet and a common-size income statement, every item is expressed as a percentage of:

a. Total equity and net income
b. Total assets and net income
c. Total assets and sales
d. Sales and total assets

34. You save $300 at the end of every month in an account paying 7.2%, compounded monthly. How many years until the balance reaches $50,000?

a. 115.87 years
b. 9.97 years
c. 13.89 years
d. 9.66 years

35. Which of the following is NOT a reason why a stock is harder to value than a bond?

a. Dividends are paid at the board's discretion, while coupons are contractual
b. A stock has no maturity date
c. Both the required return and the growth rate must be estimated
d. A stock's value is not the present value of its future cash flows

36. Why does a call provision lower a bond's value to investors?

a. Because the call premium is paid to the issuer
b. Because the issuer calls the bond when rates have fallen, exactly when holders would like to keep the high coupon, so investors demand a higher yield
c. Because callable bonds cannot be traded in the secondary market
d. Because a call provision raises the bond's default risk

37. A firm reports net income of $640,000 with 200,000 shares outstanding and pays total dividends of $240,000. What are earnings per share, dividends per share and the addition to retained earnings?

a. $3.20; $1.20; $400,000
b. $3.20; $1.20; $640,000
c. $2.00; $1.20; $400,000
d. $3.20; $0.375; $400,000

38. Expected inflation rises, while nothing changes about an issuer's creditworthiness or the real return investors require. What happens to the issuer's outstanding fixed-coupon bonds?

a. Their prices rise because nominal coupons are worth more
b. Their prices are unaffected because default risk is unchanged
c. Nominal yields rise with expected inflation (Fisher effect), so their prices fall
d. Their coupon rates are reset upward

39. A company leases equipment worth $60,000. Payments are due at the beginning of each month for 3 years, and the lessor charges 9%, compounded monthly. What is the monthly lease payment?

a. $1,907.98
b. $5,654.10
c. $1,893.78
d. $1,666.67

40. Which rating is the lowest investment-grade rating?

a. Aa / AA
b. A / A
c. Baa3 / BBB−
d. Ba1 / BB+

41. Which statement is NOT one of the cautions about financial ratio analysis?

a. Firms may use different accounting methods such as FIFO or LIFO
b. Different fiscal year-ends mean balance sheets are snapshots on different dates
c. Each industry has a single theoretically correct value for every ratio
d. Firms may window-dress items around reporting dates

42. You want $1,000,000 in 25 years. Your account pays 6%, compounded monthly. How much must you deposit at the end of every month?

a. $1,518.89
b. $1,435.83
c. $1,443.01
d. $3,333.33

43. In the dividend growth model $P_0 = D_1 / (R - g)$, what happens if the assumed growth rate g is greater than or equal to R?

a. The share is worth an infinite amount, which is a realistic result for growth firms
b. The formula breaks down (negative or infinite price), signalling that such growth cannot last forever
c. The model still works if $D_0$ is used instead of $D_1$
d. The price equals $D_1 / g$

44. Which statement about tax rates is correct?

a. The average tax rate is the one used for investment decisions
b. Under the U.S. flat 21% corporate tax, the marginal and average rates are equal at every income level
c. Under a progressive schedule the average rate exceeds the marginal rate
d. The marginal rate is total taxes divided by taxable income

45. A firm has a total debt ratio of 0.60. What are its debt-equity ratio and equity multiplier?
a. D/E 1.50; EM 2.50
b. D/E 0.60; EM 1.60
c. D/E 2.50; EM 1.50
d. D/E 1.50; EM 1.67

46. You deposit $500 at the beginning of every month for 15 years. The account pays 6%, compounded monthly. What will you have at the end of 15 years?
a. $146,136.40
b. $145,409.36
c. $148,035.17
d. $90,000.00

47. In a nonconstant-growth valuation, dividends become constant-growth after date t, and the terminal price $P_t = D_{t+1} / (R - g)$ is computed. By how many periods should $P_t$ be discounted to get today's value?
a. t + 1 periods, because it is based on $D_{t+1}$
b. t periods, because $P_t$ is a price at date t
c. One period, because the model already discounts the dividends
d. It should not be discounted; it is added to $P_0$ directly

48. Which of the following is a positive covenant in a bond indenture?
a. Limit the dividends the firm may pay
b. Prohibit mergers without creditor consent
c. Maintain working capital above a stated level
d. Forbid the issue of additional long-term debt

49. Which statement is correct?
a. The firm receives cash every time its shares trade on the NYSE
b. In the secondary market investors trade existing shares and the firm receives nothing
c. In the primary market investors trade shares among themselves
d. An IPO is a secondary-market transaction

50. Why does a balance grow faster under compound interest than under simple interest?
a. Because the interest rate rises each year
b. Because interest is earned on previously earned interest as well as on the principal
c. Because simple interest is paid only at maturity
d. Because compounding uses the APR instead of the EAR

51. Which component of a corporate bond's nominal yield is bond-specific rather than part of the term structure?
a. Real rate
b. Inflation premium
c. Interest rate risk premium
d. Default risk premium

52. You deposit $1,000 at the beginning of every quarter into an account paying 8%, compounded quarterly. How many years until the balance reaches $100,000?

a. 54.81 years
b. 13.70 years
c. 13.87 years
d. 25.00 years

53. Coca-Cola and PepsiCo both earn an ROE of about 40%. According to the DuPont breakdown in the lecture, which statement is correct?

a. Coca-Cola gets there through a high profit margin; PepsiCo relies on faster asset turnover and greater financial leverage
b. PepsiCo gets there through a high profit margin; Coca-Cola relies on leverage
c. Both firms have nearly identical margins, turnover and leverage
d. Coca-Cola has the higher asset turnover because it owns more plants

54. You want to earn an effective annual rate of 8% with monthly compounding. What APR must you be quoted?

a. 8.00%
b. 7.72%
c. 8.30%
d. 0.64%

55. A dealer quotes a stock at bid $49.90 / ask $50.10. Which statement is correct?

a. You can sell to the dealer at $49.90 and buy from the dealer at $50.10; the $0.20 spread is the dealer's compensation
b. You can buy at $49.90 and sell at $50.10
c. A broker would quote the same prices and hold the stock in inventory
d. The spread is a commission the dealer pays to the exchange

56. Why does a bond buyer pay the seller accrued interest on top of the quoted (clean) price?

a. Because the dealer adds a commission to every trade
b. Because the next coupon is paid in full to whoever holds the bond on the payment date, so the buyer reimburses the seller for the part the seller earned
c. Because accrued interest is a tax collected at settlement
d. Because the clean price already includes the next coupon

57. A firm reports negative cash flow from assets for the year. Which interpretation is most appropriate?

a. The firm is necessarily in financial distress
b. The firm must have reported a net loss
c. The firm raised funds externally during the year, possibly to finance growth
d. The firm's cash flow to creditors must be negative

58. In a loan repaid with equal payments, how does each payment change over time?
a. The interest portion rises and the principal portion falls
b. The payment is constant, the interest portion falls and the principal portion rises
c. Both the interest and principal portions are constant
d. The payment falls each period as the balance falls

59. Which statement about NASDAQ is correct?
a. It is an auction market organized around designated market makers on a trading floor
b. It is a dealer market: a computer network in which several market makers compete in the same stock
c. It has no market makers; all trades are matched between customers
d. It lists only preferred stock

60. Why is inventory subtracted from current assets in the quick (acid-test) ratio?
a. Because inventory is not a current asset
b. Because inventory is the least liquid current asset and its book value is the least reliable
c. Because inventory is already counted in cash
d. Because inventory is financed with long-term debt

61. Which of the following is NOT one of the assumptions behind a bond's yield to maturity?
a. The bond is held to maturity
b. There is no default
c. Coupons are reinvested at the YTM
d. Coupons are reinvested at the coupon rate

62. You deposited $200 at the end of every month for 10 years, and the account is now worth $40,000. What annual rate, compounded monthly, did it earn?
a. 9.58%
b. 0.80%
c. 10.01%
d. 6.67%

63. Which statement about the tax treatment of financing costs is correct?
a. Both interest and dividends are tax-deductible to the paying corporation
b. Interest is tax-deductible to the corporation; dividends are not
c. Dividends are tax-deductible to the corporation; interest is not
d. Neither affects the corporation's tax bill

64. Your budget allows a car payment of $520 per month for 5 years. The bank charges 9.6%, compounded monthly. How large a loan can you take?
a. $5,394.53
b. $2,538.75
c. $31,200.00
d. $24,702.19

65. A firm skipped two years of dividends on its cumulative preferred stock. Which statement is correct?

a. The firm is in default and the preferred holders can force bankruptcy
b. The arrears must be paid before any dividend is paid on common stock
c. The skipped dividends are forgiven when the board resumes payments
d. Preferred holders immediately become creditors

66. A loan is quoted at 12%, compounded quarterly. What is the effective annual rate?

a. 12.55%
b. 12.00%
c. 3.00%
d. 12.68%

67. PepsiCo's current ratio is 0.85. According to the lecture, how should this be read?

a. As a clear sign of imminent financial distress
b. Not necessarily a red flag: steady cash sales fund working capital, long supplier terms inflate payables, and strong credit lets it roll short-term debt
c. As evidence that PepsiCo holds too much idle cash
d. As proof that PepsiCo's ROE must be lower than Coca-Cola's

68. What is the value of a firm that is not paying dividends today?

a. Zero, because the dividend discount model only values dividends already paid
b. It has value only because it is expected to pay dividends (or be bought by someone who will) at some point in the future
c. It equals its book value per share
d. It can only be valued with the P/E ratio of the same firm

69. In the short run, which of the following is best described as a fixed cost?

a. Raw materials consumed in production
b. Sales commissions paid per unit sold
c. A lease payment on a warehouse under a five-year contract
d. Hourly wages of temporary production workers

70. How does a put provision affect a bond compared with an otherwise identical bond without one?

a. It lowers the bond's value because the issuer can buy it back early
b. It raises the bond's value because the holder may force the issuer to buy it back at a stated price
c. It has no effect on value
d. It converts the bond into equity

# Business Finance Midterm — Answers

**1. b** [Exhibit A]
**EPS** = 948 / 100 = 9.48, so P/E = 74 / 9.48 = 7.81
**sales per share** = 4,800 / 100 = 48, so P/S = 74 / 48 = 1.54
**book value per share** = 3,700 / 100 = 37, so M/B = 74 / 37 = 2.00.

**2. b** [Exhibit A]
**TIE** = EBIT / interest = 1,400 / 200 = 7.00
**cash coverage** = (EBIT + depreciation) / interest = 1,900 / 200 = 9.50.

**3. b** [Exhibit A]
Common-size balance sheet items are divided by total assets: 900 / 6,400 = 14.06%
common-size income statement items are divided by sales: 948 / 4,800 = 19.75%.

**4. a** [Exhibit A]
**EV** = 7,400 + 2,700 − 200 = 9,900 (total liabilities = 6,400 − 3,700 = 2,700)
**EBITDA** = 1,400 + 500 = 1,900
**EV/EBITDA** = 9,900 / 1,900 = 5.21.

**5. a** [Exhibit A]
**EPS** = 948 / 100 = 9.48
**price** = EPS × benchmark P/E = 9.48 × 12 = $113.76
**BA II Plus:** 948 ÷ 100 = 9.48, × 12 = 113.76.

**6. a** [Exhibit A]
**Inventory turnover** = COGS / inventory = 2,900 / 900 = 3.22
**receivables turnover** = sales / AR = 4,800 / 700 = 6.86
**total asset turnover** = sales / total assets = 4,800 / 6,400 = 0.75.

**7. d** [Exhibit A]
Accounts receivable rose by 100 (an increase in an asset is a use of cash)
accounts payable rose by 40 (an increase in a liability is a source of cash)
**net effect** = −100 + 40 = $60 use of cash.

**8. b**
The coupon rate (annual coupon ÷ face value) is fixed when the bond is issued
the YTM is the market's required return and changes daily, so the price is whatever makes the two consistent.

**9. a**
GAAP lists the most liquid assets first and the liabilities due soonest first
IFRS / K-IFRS flips the order and lists non-current items first, with the same financial logic.

**10. b**
**Market value of equity** = 20 million × $12 = $240 million versus book value of $80 million, a difference of $160 million
book values rest on historical cost, so a gap is normal and market value is what matters for decisions.

**11. b**
The premium is given back to the holder as a capital loss over the remaining life, so the price drifts down to par and equals the face value at maturity.

**12. b**
**r** = 0.5% per month, t = 360: 250,000 = C × [1 − 1/(1.005)$^{360}$] / 0.005, so C = $1,498.88 (360 N, 0.5 I/Y, 250000 PV, 0 FV, CPT PMT).

**13. a**
**Dividend yield** = $D_1 / P_0$ = 2.00 / 50.00 = 4.0%
under constant growth the price grows at g, so the capital gains yield = 6.0%
**R** = 4.0% + 6.0% = 10.0%
**BA II Plus:** 2 ÷ 50 = 0.04, + 0.06 = 0.10.

**14. d**
Interest rate risk is greater for a longer maturity and for a lower coupon
a zero coupon bond is the extreme case because all of its value sits at maturity, so the 30-year zero is the most rate-sensitive.

**15. d**
Preferred stock looks like debt (fixed dividend, no vote, ranks ahead of common, often callable), but missing a dividend is not a default, the dividend is not tax-deductible and holders rank behind every creditor, so it is equity.

**16. a**
An increase in NWC is a cash outflow: the firm invested cash in current assets net of current liabilities, so CFFA = OCF − NCS − ΔNWC.

**17. b**
**Fisher:** 1 + R = (1 + r)(1 + h), so r = 1.08 / 1.03 − 1 = 4.85%
**BA II Plus:** 1.08 ÷ 1.03 = 1.04854, − 1 = 0.0485.

**18. a**
**Accrued interest** = 30 × (4 / 6) = 20
**dirty price** = clean price + accrued interest = 1,012 + 20 = $1,032.00
**BA II Plus:** 30 × 4 ÷ 6 = 20, + 1012 = 1,032.

**19. a**
**r** = 0.4%, t = 240: FV = 250 × [(1.004)$^{240}$ − 1] / 0.004 = $100,418.76 (240 N, 0.4 I/Y, −250 PMT, 0 PV, CPT FV).

**20. b**
Leverage enters ROE through the equity multiplier (ROE = ROA × EM)
it magnifies returns in both directions, so it raises ROE when assets earn more than debt costs and lowers it further when they earn less.

**21. b**
**Price** = sales per share × benchmark price-sales ratio = 12.00 × 1.8 = $21.60
**BA II Plus:** 12 × 1.8 = 21.60.

**22. b**
**Equivalent taxable yield** = muni yield / (1 − t) = 4.5% / 0.75 = 6.00%
**BA II Plus:** 4.5 ÷ 0.75 = 6.00.

**23. b**
**Paid-in surplus** = (issue price − par) × shares = (15 − 1) × 5,000 = $70,000
the $5,000 of par goes to the common stock line.

**24. d**
The dividend and the rate must be on the same basis: r = 8% / 4 = 2% per quarter
**PV** = C / r = 1.25 / 0.02 = $62.50
**BA II Plus:** 1.25 ÷ 0.02 = 62.50.

**25. d**
**BGN mode:** 36 N, 36000 PV, −1150 PMT, 0 FV, CPT I/Y = 0.8227% per month
**quoted rate** = 0.8227 × 12 = 9.87%
switch BGN off afterwards
**Formula:** $36{,}000 = 1{,}150 \times [1 - 1/(1 + r)^{36}] / r \times (1 + r)$, solved by iteration.

**26. b**
**OCF** = EBIT + depreciation − taxes measures the cash generated by operations before financing
interest belongs to the cash flow to creditors.

**27. c**
More frequent compounding pays interest on interest more often, so the EAR = $(1 + \text{APR}/m)^m - 1$ rises with m (15% APR: 15.00% annual, 16.08% monthly, 16.18% daily).

**28. c**
If the coupon rate exceeds the yield, the coupon is generous in the market and investors bid the price above face value
the premium is given back as a capital loss over the remaining life.

**29. b**
**OCF** = EBIT + depreciation − taxes adds depreciation back and leaves interest out, because it measures the cash generated by operations before financing net income is an accrual figure after both.

**30. b**
**EBIT** = 850,000 − 610,000 − 95,000 = 145,000
**taxable income** = 145,000 − 32,000 = 113,000
**taxes** = 0.21 × 113,000 = 23,730
**net income** = 113,000 − 23,730 = $89,270.

**31. a**
**r** = 2% per quarter: PV = $2{,}000/1.02 + 3{,}000/(1.02)^2 + 0/(1.02)^3 + 4{,}000/(1.02)^4$ = $8,539.67
**BA II Plus:** CF, CF0 = 0, C01 = 2000, C02 = 3000, C03 = 0, C04 = 4000, NPV I = 2, CPT = 8,539.67.

**32. a**
**Current yield** = 70 / 1,075 = 6.51%
for a premium bond the ordering is the reverse of a discount bond: YTM (5.98%) < current yield (6.51%) < coupon rate (7.00%)
**BA II Plus:** 70 ÷ 1075 = 0.0651
the YTM comes from 10 N, −1075 PV, 70 PMT, 1000 FV, CPT I/Y = 5.98.

**33. c**
Common-size balance sheets divide every account by total assets
common-size income statements divide every line by sales, which allows comparison over time and across firms of different sizes.

**34. d**
**r** = 0.6%: 0.6 I/Y, −300 PMT, 0 PV, 50000 FV, CPT N = 115.87 months
115.87 / 12 = 9.66 years
**Formula:** $50{,}000 = 300 \times [(1.006)^t - 1] / 0.006$, t in months.

**35. d**
The discounting principle is identical for stocks and bonds: the value of any financial asset is the PV of the cash flows it delivers
what is harder is the inputs (discretionary dividends, no maturity, estimated R and g).

**36. b**
The call option belongs to the issuer, who uses it when rates fall
investors are left reinvesting at lower rates, so they require a higher yield (a lower price) on a callable bond. A put provision is the mirror image and raises a bond's value.

**37. a**
**EPS** = 640,000 / 200,000 = $3.20
**DPS** = 240,000 / 200,000 = $1.20
**addition to retained earnings** = 640,000 − 240,000 = $400,000.

**38. c**
**Bond yields are nominal:** 1 + R = (1 + r)(1 + h), so a higher h raises the required nominal yield and lowers the price of a fixed-coupon bond.

**39. c**
r = 0.75%, t = 36: 60,000 = C × [1 − 1/(1.0075)$^{36}$] / 0.0075 × 1.0075, so C = $1,893.78 (BGN mode: 36 N, 0.75 I/Y, 60000 PV, 0 FV, CPT PMT
switch BGN off afterwards).

**40. c**
Investment grade runs from Aaa/AAA down to Baa3/BBB−
anything below is speculative grade (junk). Ratings measure default risk, not interest rate risk.

**41. c**
There is no underlying theory giving a single correct value
a ratio needs a benchmark (time-trend or peer-group analysis), and comparisons can be distorted by accounting methods, fiscal year-ends, conglomerates, capital structure, seasonality and window dressing.

**42. c**
r = 0.5%, t = 300: 1,000,000 = C × [(1.005)$^{300}$ − 1] / 0.005, so C = $1,443.01 (300 N, 0.5 I/Y, 0 PV, 1000000 FV, CPT PMT).

**43. b**
The model requires R > g
if g ≥ R the algebra returns a negative or infinite value, which means the growth assumption is impossible to sustain indefinitely. The g in the model is the rate the firm can sustain forever.

**44. b**
The marginal rate (on the next dollar) is used for decisions
**with a flat rate marginal** = average, and with a progressive schedule (e.g. Korea) the marginal rate is above the average rate.

**45. a**
**Equity is 0.40 of assets:** EM = TA / TE = 1 / 0.40 = 2.50
**D/E** = 0.60 / 0.40 = 1.50, and EM = 1 + D/E.

**46. a**
r = 0.5%, t = 180: ordinary FV = 500 × [(1.005)$^{180}$ − 1] / 0.005 = 145,409.36
**annuity due** = 145,409.36 × 1.005 = $146,136.40
**BA II Plus:** 180 N, 0.5 I/Y, −500 PMT, 0 PV, CPT FV = 145,409.36, × 1.005 = 146,136.40 (or BGN mode switch BGN off afterwards).

**47. b**
$P_t$ is the value at date t of all dividends from t + 1 onward, so it is discounted by $(1 + R)^t$
discounting by t + 1 is the most common mistake.

**48. c**
Positive covenants state what the firm must do (maintain working capital or ratios, furnish audited statements, keep collateral in good condition)
the others are negative covenants. Firms accept covenants because the lower yield is worth more than the lost flexibility.

**49. b**
In the primary market the issuer sells securities for the first time and raises money
in the secondary market existing shares change hands between investors, and the firm receives nothing.

**50. b**
Simple interest is paid only on the original principal
compound interest pays interest on principal and accumulated interest, so FV = PV × $(1 + r)^t$ grows faster and faster.

**51. d**
The term structure (real rate + inflation premium + interest rate risk premium) is the same for every bond of a given maturity
the default risk, taxability and liquidity premiums are specific to each bond.

**52. b**
r = 2% per quarter, BGN mode: 2 I/Y, −1000 PMT, 0 PV, 100000 FV, CPT N = 54.81 quarters
54.81 / 4 = 13.70 years
switch BGN off afterwards
**Formula:** 100,000 = 1,000 × [(1.02)$^t$ − 1] / 0.02 × 1.02, t in quarters.

**53. a**
**KO:** profit margin 27.34%, asset turnover 0.46, equity multiplier 3.26
**PEP:** 8.77%, 0.87, 5.26. Same ROE, different routes: KO's asset-light concentrate model gives high margins.

**54. b**
**APR** = 12 × [(1.08)$^{1/12}$ − 1] = 12 × 0.6434% = 7.72%
(ICONV: EFF = 8, C/Y = 12, CPT NOM).

**55. a**
The bid is the price the dealer pays (you sell), the ask is the price the dealer accepts (you buy), and the spread compensates the dealer for making a market and bearing inventory risk
a broker only matches trades for a commission.

**56. b**
**The dirty (invoice) price** = clean price + accrued interest, where accrued interest = coupon × days since last coupon ÷ days in the coupon period.

**57. c**
Negative CFFA means operations plus asset purchases used more cash than they generated, so the gap was financed by new debt or equity
this is routine for fast-growing firms making heavy investments.

**58. b**
**Interest each period** = rate × beginning balance
as the balance falls, interest falls and, with a constant payment, the principal portion rises.

**59. b**
NASDAQ has no physical floor
trades are executed against competing market makers, whose competition narrows the spread. The NYSE is a hybrid auction market with DMMs, now largely electronic.

**60. b**
**Quick ratio** = (current assets − inventory) / current liabilities
inventory may not be sold quickly at book value, especially when a firm is in trouble.

**61. d**
YTM is the bond's internal rate of return
it is a promised yield that assumes the bond is held to maturity, no default occurs and coupons are reinvested at the YTM itself.

**62. a**
120 N, −200 PMT, 0 PV, 40000 FV, CPT I/Y = 0.7984% per month
**quoted rate** = 0.7984 × 12 = 9.58% (the EAR would be 10.01%)
**Formula:** $40{,}000 = 200 \times [(1 + r)^{120} - 1] / r$, solved by iteration.

**63. b**
Interest is deducted before computing taxable income
dividends are paid out of after-tax profit, which gives debt a tax advantage at the corporate level.

**64. d**
**r** = 9.6% / 12 = 0.8% per month, t = 60: PV = $520 \times [1 - 1/(1.008)^{60}] / 0.008$ = $24,702.19 (60 N, 0.8 I/Y, −520 PMT, 0 FV, CPT PV).

**65. b**
Cumulative preferred dividends accumulate as arrearages
no common dividend can be paid until they are cleared, but skipping them is not a default.

**66. a**
**EAR** = $(1 + 0.12/4)^4 - 1 = (1.03)^4 - 1$ = 12.55% (ICONV: NOM = 12, C/Y = 4, CPT EFF).

**67. b**
A current ratio below 1 can be normal for a firm with steady cash sales, large payables and cheap access to short-term credit
a big current-asset cushion would only sit idle.

**68. b**
The price of a share is the PV of all future dividends
a firm paying nothing now must be expected to pay eventually, otherwise its shares are worthless.

**69. c**
A fixed cost does not vary with output over the short-run horizon
a contractual lease payment must be paid whatever the output level.

**70. b**
A put gives the option to the holder, the mirror image of a call provision, so it raises the bond's value (lowers its yield).

# Business Finance Midterm

Written Calculations · Chapters 2–7 · 15 questions

## Exhibit A · Harborview Outfitters, Inc.

*Used in questions 1–3*

Balance sheets, December 31 ($ millions)

| Item | 2024 | 2025 |
|---|---|---|
| Assets | | |
| Cash | 150 | 200 |
| Accounts receivable | 600 | 700 |
| Inventory | 850 | 900 |
| Total current assets | 1,600 | 1,800 |
| Net fixed assets | 4,400 | 4,600 |
| Total assets | 6,000 | 6,400 |
| Liabilities and owners' equity | | |
| Accounts payable | 520 | 560 |
| Notes payable | 280 | 240 |
| Total current liabilities | 800 | 800 |
| Long-term debt | 1,800 | 1,900 |
| Common stock and paid-in surplus | 1,400 | 1,500 |
| Retained earnings | 2,000 | 2,200 |
| Total owners' equity | 3,400 | 3,700 |
| Total liabilities and owners' equity | 6,000 | 6,400 |

2025 income statement ($ millions)

| Item | 2025 |
|---|---|
| Net sales | 4,800 |
| Cost of goods sold | 2,900 |
| Depreciation | 500 |
| Earnings before interest and taxes | 1,400 |
| Interest paid | 200 |
| Taxable income | 1,200 |
| Taxes (21%) | 252 |
| Net income | 948 |
| Dividends | 748 |
| Addition to retained earnings | 200 |

1. [Exhibit A] Harborview Outfitters, Inc. ($ millions). Compute the 2025 cash flow from assets in two ways: (a) from the asset side, CFFA = OCF − NCS − ΔNWC; (b) from the financing side, CFFA = CF to creditors + CF to stockholders, where net new borrowing is the change in long-term debt and net new equity is the change in common stock and paid-in surplus. Write each formula and show that both give the same figure.

2. [Exhibit A] Harborview Outfitters, Inc. ($ millions). Using 2025 year-end figures, compute the profit margin, total asset turnover and equity multiplier, combine them into ROE with the DuPont identity, and confirm the result with NI / TE.

3. [Exhibit A] Harborview Outfitters, Inc. ($ millions). For 2025, compute the current ratio, the quick ratio, the total debt ratio and the debt-equity ratio. Write each formula.

4. Assume the tax schedule in the table. A taxpayer reports taxable income of $200,000. Compute the total tax, the average tax rate and the marginal tax rate.

Tax schedule

| Taxable income | Rate |
|---|---|
| Up to $50,000 | 10% |
| $50,001 – $150,000 | 20% |
| Above $150,000 | 30% |

5. An investment of $12,000 today will be worth $20,000 in 7 years. (a) What annual rate does it earn? (b) At that rate, how many years would it take $12,000 to grow to $30,000? Write the equation for each part before solving.

6. You deposit $3,000 today and another $5,000 in 18 months. The account pays 6%, compounded monthly. How much is in the account at the end of 4 years? Show the value of each deposit separately.

7. You borrow $12,000 for 3 years at 7.2%, compounded monthly, repaid in equal monthly payments. Compute (a) the monthly payment, (b) the interest in the first payment and (c) the loan balance after the 12th payment.

8. A two-year office lease requires $1,200 at the beginning of every month. The relevant rate is 7.2%, compounded monthly. Value it first as an ordinary annuity and then as the annuity due it actually is.

9. A $22,000 car can be financed with 48 monthly payments of $540. Write the equation for the monthly rate, then report the monthly rate, the quoted annual rate (APR) and the effective annual rate.

10. Lender A charges 11.8%, compounded monthly. Lender B charges 12%, compounded semiannually. Convert both quotes to effective annual rates and state which loan is cheaper.

11. A $1,000 bond pays a 6% coupon semiannually and matures in 12 years. Price it when comparable bonds yield 7% and when they yield 5% (quoted annual rates), and say whether it sells at a premium or a discount in each case.

12. A $1,000 bond pays a 5% coupon semiannually, matures in 8 years and trades at $940. Write the pricing equation, then find the six-month yield, the quoted YTM, the effective annual yield and the current yield, and rank the coupon rate, current yield and YTM.

13. A company just paid a dividend of $1.80 per share. Dividends grow at 4% a year forever and investors require 12%. Compute (a) today's price, (b) the expected price three years from today and (c) the dividend yield and capital gains yield.

14. A firm just paid a $1.50 dividend. Dividends will grow at 15% for the next two years and at 4% forever after that. The required return is 12%. Write out each dividend, the terminal price and the value today.

15. A firm will pay no dividend for the next three years. It will pay $1.00 at the end of year 4, and that dividend will then grow at 5% forever. The required return is 15%. Value the stock today, showing the date at which you apply the growth model.

# Business Finance Midterm — Model answers

**1.** Ch.2 · Cash flow from assets, two ways
**OCF** = EBIT + depreciation − taxes = 1,400 + 500 − 252 = 1,648
**NCS** = ending net FA − beginning net FA + depreciation = 4,600 − 4,400 + 500 = 700
**ΔNWC** = (1,800 − 800) − (1,600 − 800) = 1,000 − 800 = 200
**CFFA** = OCF − NCS − ΔNWC = 1,648 − 700 − 200 = 748
**CF to creditors** = interest paid − net new borrowing = 200 − (1,900 − 1,800) = 100
**CF to stockholders** = dividends paid − net new equity = 748 − (1,500 − 1,400) = 648
**Check** = 100 + 648 = 748, the same as the asset-side figure
*Key results: 1,648 · 700 · 748 · 100 · 648*

**2.** Ch.3 · DuPont identity
**PM** = NI / sales = 948 / 4,800 = 19.75%
**TAT** = sales / TA = 4,800 / 6,400 = 0.75
**EM** = TA / TE = 6,400 / 3,700 = 1.73
**ROE** = PM × TAT × EM = 0.1975 × 0.75 × 1.73 = 25.62%
**Check** = NI / TE = 948 / 3,700 = 25.62%
*Key results: 19.75 · 0.75 · 1.73 · 25.62*

**3.** Ch.3 · Liquidity and leverage
**Current ratio** = CA / CL = 1,800 / 800 = 2.25
**Quick ratio** = (CA − inventory) / CL = (1,800 − 900) / 800 = 1.13
**Total debt** = TA − TE = 6,400 − 3,700 = 2,700
**Total debt ratio** = TD / TA = 2,700 / 6,400 = 0.42
**D/E** = TD / TE = 2,700 / 3,700 = 0.73
*Key results: 2.25 · 1.13 · 0.42 · 0.73*

**4.** Ch.2 · Average and marginal tax rates
**Tax** = 0.10 × 50,000 + 0.20 × 100,000 + 0.30 × 50,000 = 5,000 + 20,000 + 15,000 = 40,000
**Average rate** = total tax / taxable income = 40,000 / 200,000 = 20%
**Marginal rate** = rate on the next dollar of income = 30%
*Key results: 40,000 · 20% · 30%*

**5.** Ch.4 · Rate and time
**(a) Equation:** $20{,}000 = 12{,}000 \times (1 + r)^{7}$
**r** = $(20{,}000 / 12{,}000)^{1/7} - 1 = 7.57\%$
**BA II Plus:** 7 N, −12000 PV, 0 PMT, 20000 FV, CPT I/Y = 7.57
**(b) Equation:** $30{,}000 = 12{,}000 \times (1.0757)^{t}$
t = ln(2.5) / ln(1.0757) = 12.56 years
**BA II Plus:** 7.57 I/Y, −12000 PV, 0 PMT, 30000 FV, CPT N = 12.56
*Key results: 7.57 · 12.56*

**6.** Ch.5 · Two deposits
**r** = 6% / 12 = 0.5% per month
**FV** = $3{,}000 \times (1.005)^{48} + 5{,}000 \times (1.005)^{30}$
**First deposit** = $3{,}000 \times (1.005)^{48} = 3{,}811.47$ (48 months)
**Second deposit** = $5{,}000 \times (1.005)^{30} = 5{,}807.00$ (30 months)
**Total** = 9,618.47
**BA II Plus:** 48 N, 0.5 I/Y, 0 PMT, −3000 PV, CPT FV = 3,811.47
30 N, −5000 PV, CPT FV = 5,807.00
*Key results: 3,811.47 · 5,807.00 · 9,618.47*

**7.** Ch.5 · Amortizing loan
**r** = 7.2% / 12 = 0.6% per month, N = 36
**Payment:** 12,000 = C × [1 − 1/(1.006)$^{36}$] / 0.006, so C = 371.62
**First interest** = 0.006 × 12,000 = 72.00
**Balance after 12** = PV of the 24 remaining payments = 371.62 × [1 − 1/(1.006)$^{24}$] / 0.006 = 8,283.46
**BA II Plus:** 36 N, 0.6 I/Y, 12000 PV, 0 FV, CPT PMT = −371.62
**AMORT:** P1 = 1, P2 = 12, BAL = 8,283.46
*Key results: 371.62 · 72.00 · 8,283.46*

**8.** Ch.5 · Lease value (annuity due)
**r** = 0.6% per month, N = 24
**Ordinary annuity** = 1,200 × [1 − 1/(1.006)$^{24}$] / 0.006 = 26,747.92
**Annuity due** = ordinary annuity × (1 + r) = 26,747.92 × 1.006 = 26,908.41
**BA II Plus:** 24 N, 0.6 I/Y, −1200 PMT, 0 FV, CPT PV = 26,747.92, then × 1.006 (or BGN mode, switched off afterwards)
*Key results: 26,747.92 · 26,908.41*

**9.** Ch.5 · Rate on a car loan
**Formula:** 22,000 = 540 × [1 − 1/(1 + r)$^{48}$] / r, solved by iteration
**Monthly rate** = 0.6902%
**APR** = 0.6902% × 12 = 8.28%
**EAR** = (1.006902)$^{12}$ − 1 = 8.60%
**BA II Plus:** 48 N, 22000 PV, −540 PMT, 0 FV, CPT I/Y = 0.6902
**ICONV:** NOM = 8.28, C/Y = 12, CPT EFF = 8.60
*Key results: 0.69 · 8.28 · 8.60*

**10.** Ch.5 · Comparing lenders
**EAR(A)** = (1 + 0.118/12)$^{12}$ − 1 = 12.46%
**EAR(B)** = (1 + 0.12/2)$^{2}$ − 1 = 12.36%
**Decision:** borrow from Lender B, which has the lower EAR
**ICONV:** NOM = 11.8, C/Y = 12, CPT EFF = 12.46
**NOM** = 12, C/Y = 2, CPT EFF = 12.36
*Key results: 12.46 · 12.36 · Lender B*

**11.** Ch.6 · Bond pricing
**Inputs:** N = 12 × 2 = 24, PMT = 60 / 2 = 30, FV = 1,000
**At 7%:** P = 30 × [1 − 1/(1.035)$^{24}$] / 0.035 + 1,000 / (1.035)$^{24}$ = 919.71, a discount (coupon rate < yield)
**At 5%:** P = 30 × [1 − 1/(1.025)$^{24}$] / 0.025 + 1,000 / (1.025)$^{24}$ = 1,089.42, a premium (coupon rate > yield)
**BA II Plus:** 24 N, 3.5 I/Y, 30 PMT, 1000 FV, CPT PV = −919.71
2.5 I/Y, CPT PV = −1,089.42
*Key results: 919.71 · 1,089.42 · discount · premium*

**12.** Ch.6 · Yield to maturity
**Formula:** 940 = 25 × [1 − 1/(1 + r)$^{16}$] / r + 1,000 / (1 + r)$^{16}$, solved by iteration
**Six-month yield** = 2.98%
**Quoted YTM** = 2.98% × 2 = 5.95%
**Effective annual yield** = (1.0298)$^{2}$ − 1 = 6.04%
**Current yield** = annual coupon / price = 50 / 940 = 5.32%
**Ranking:** coupon rate 5.00% < current yield 5.32% < YTM 5.95% (a discount bond)
**BA II Plus:** 16 N, −940 PV, 25 PMT, 1000 FV, CPT I/Y = 2.98
*Key results: 2.98 · 5.95 · 6.04 · 5.32*

**13.** Ch.7 · Constant growth
$D_1 = D_0 \times (1 + g) = 1.80 \times 1.04 = 1.872$
$P_0 = D_1 / (R - g) = 1.872 / 0.08 = 23.40$
$P_3 = D_4 / (R - g) = 1.80 \times (1.04)^4 / 0.08 = 2.1057 / 0.08 = 26.32$
**Dividend yield** = $D_1 / P_0 = 1.872 / 23.40 = 8\%$
**Capital gains yield** = g = 4% (R = 8% + 4% = 12%)
*Key results: 23.40 · 26.32 · 8% · 4%*

**14.** Ch.7 · Nonconstant growth
$D_1 = 1.50 \times 1.15 = 1.725$
$D_2 = 1.725 \times 1.15 = 1.98375$
$D_3 = 1.98375 \times 1.04 = 2.0631$
$P_2 = D_3 / (R - g) = 2.0631 / (0.12 - 0.04) = 25.79$
$P_0 = 1.725 / 1.12 + (1.98375 + 25.79) / (1.12)^2 = 1.540 + 22.140 = 23.68$
**BA II Plus CF:** CF0 = 0, C01 = 1.725, C02 = 27.773, NPV I = 12, CPT = 23.68
*Key results: 1.725 · 1.98375 · 2.0631 · 25.79 · 23.68*

**15.** Ch.7 · A firm that pays nothing yet
Growth model at date 3, one period before the first dividend: $P_3 = D_4 / (R - g) = 1.00 / (0.15 - 0.05) = 10.00$
$P_0 = P_3 / (1.15)^3 = 10.00 / 1.5209 = 6.58$
**BA II Plus:** 3 N, 15 I/Y, 0 PMT, 10 FV, CPT PV = −6.58
*Key results: 10.00 · 6.58*